

DISCOVERED THAT  
**2, 8, 20,  
28, 50,  
82, OR 126**  
**PROTONS AND  
NEUTRONS**  
**MAKE THE  
NUCLEUS  
STABLE**



**FINALLY  
APPOINTED  
PROFESSOR  
OF PHYSICS  
AT THE AGE  
OF 54**

**1 OF JUST  
3 FEMALE  
NOBEL PRIZE-WINNING  
PHYSICISTS**

to join the Manhattan Project, which was building the world's first atomic weapon. She joined a group of scientists at Columbia University who were seeking to separate the isotope U-235 from natural uranium in their research toward building an atomic bomb.

#### **Nuclear shell model**

In the late 1940s, Goeppert Mayer was working at the Argonne National Laboratory and the Institute for Nuclear Studies in Chicago. She was studying why some isotopes (different forms of the same element) with certain numbers of protons and neutrons in their nucleus were more stable than others and were therefore more likely to exist. She had found seven such numbers. When the amount of protons or neutrons in the nucleus was one of these seven numbers, the isotope would have a particularly stable nuclear structure, but she could not fully explain why. In 1949, the nuclear physicist Enrico Fermi (see p.196) asked her: "Is there any indication of spin-orbit coupling?" Goeppert Mayer said this question gave her the answer she was

looking for, and the next day she put forward her theory of the nuclear shell model. Electrons were known to exhibit spin-orbit coupling—they orbited the nucleus in pairs, with one spinning clockwise and the other one counterclockwise. Goeppert Mayer reasoned that pairs of protons and neutrons were behaving in the same way in concentric shell-like structures within the nucleus. She described these pairs as being similar to circles of dancers waltzing in different directions and spinning at the same time.

In her theory, Goeppert Mayer stated that when the outer shell of the nucleus was full (containing one of her seven numbers of protons or neutrons), it would be very stable. When it was not full, the nucleus would be less stable. This helps explain why elements such as helium, oxygen, and calcium are common: they have stable nuclei that fit this number pattern. Goeppert Mayer's mathematical background and

#### **EUGENE WIGNER**

**US professor of mathematical physics Eugene Wigner furthered the theory of the atomic nucleus and elementary particles, for which he shared the Nobel Prize in Physics in 1963.**



Wigner (1902–1995) worked at the University of Göttingen. In 1933, he worked out that the binding force of nucleons (protons and neutrons) depends on how close they are together. He is thought to have coined the term "magic numbers" for the stable numbers of nucleons described by Goeppert Mayer. In 1939, Wigner and others encouraged Einstein to send a letter to President Roosevelt, which led to the Manhattan Project.

